

LANDSLIDENEI — Phase 8N Technical Walkthrough

Windows Desktop Application Packaging, Standalone Installer & Website Release Integration

Document ID	Release Stage	Date	Platform Architecture	Installer Artifact
DOC-PHASE-8N-APP-PKG	Phase 8N (Frozen)	September 2026	Windows 10/11 x64	LANDSLIDENEI_Setup_x64.exe

1. Executive Summary & Scope Constraints

Phase 8N establishes the production-grade delivery mechanism for LANDSLIDENEI by converting the operational Python/FastAPI geospatial workstation into a standalone, 100% offline Windows desktop application with an automated self-extracting installer. All scientific assets, machine learning models, and API schemas were strictly preserved without modification.

Constraint / Invariant	Requirement Specification	Verification Result
Zero Model Drift	Frozen Model A pipeline and metadata must remain bitwise identical	VERIFIED (10/10 Match)
Zero Python Dependency	Runs on standard Windows 10/11 PCs without Python or git installed	PASSED (Embedded Runtime)
100% Offline Capability	Full GIS map, risk profiler, and boundary GeoJSON load locally	PASSED (Zero Cloud Leaks)
Website Download Link	Download button must trigger official GitHub release installer asset	CONNECTED (v1.0.0 Link)
No Git Binary Bloat	.gitignore must exclude dist/, build/, and installer/*.exe	VERIFIED (Rule Active)

2. Desktop Application Architecture & Lifecycle

The standalone desktop application (desktop_app.py) wraps the proven FastAPI risk engine and Leaflet GIS workstation within a native Windows desktop experience using Microsoft Edge dedicated application mode (--app):

- Dual-Mode Root Resolution:** Dynamically resolves APP_ROOT from sys.executable parent or PyInstaller _MEIPASS, setting LANDSLIDENEI_ROOT environment variable across all inference submodules.
- Dedicated Backend Worker Thread:** Runs Uvicorn asynchronously on a secondary thread with its own isolated event loop, keeping the main thread responsive for Windows process lifecycle management.
- Dynamic Port Negotiation:** Automatically scans starting from port 8000 for an available socket, eliminating conflicts with existing local services.
- Headless Health Check CLI:** Implements --check-health flag enabling automated CI/CD and verification test suites to audit HTTP 200 readiness without launching GUI windows.
- One-Click Shortcut Helpers:** Bundles Create_Shortcuts.bat and Create_Desktop_Shortcut.vbs for automatic creation of Desktop and Start Menu icons pointing to assets/icon.ico.

Runtime Guarantee: When executed, LANDSLIDENEI binds to 127.0.0.1, verifies model integrity, polls its internal health endpoint, launches the workstation UI in a standalone borderless frame, and gracefully shuts down all server threads when the window is closed.

3. Packaging Optimization & Hardening

During PyInstaller compilation, several Windows-specific packaging hurdles were systematically identified and resolved:

Engineering Challenge	Root Cause Analysis	Implemented Engineering Solution
PyTorch Footprint Bloat	Auto-discovery collected PyTorch CUDA libs (~2 GB total size)	Explicitly excluded torch, torchvision, torchaudio (slashed size to 451 MB)
Test Suite Analysis Overhead	Collecting shapely, rasterio, scipy included thousands of test files	Added --exclude-module for shapely.tests, rasterio.tests, etc. (70% faster build)
Windowed Stream NoneType	PyInstaller --windowed sets sys.stdout/stderr to None, causing logger crash	Added io.StringIO() fallback + Uvicorn log_config=None + preserved ORIG_STDOUT
Antivirus Cold-Start Lag	Windows Defender deep-scans bundled C-extensions on first unpack	Increased startup polling timeout to 60.0s with diagnostic heartbeat logging

4. Distribution Workflow & Website Integration

The public landing page and installer delivery pipeline are unified into an enterprise distribution workflow:

- Direct Installer CTA:** The 'Download for Windows' button on website/index.html triggers downloadReleasePackage() in website/js/app.js, initiating download of LANDSLIDENEI_Setup_x64.exe.
- GitHub Actions Release Pipeline:** Created .github/workflows/windows-build.yml to automatically checkout repo, install dependencies, compile standalone binary, package 7-Zip SFX, and publish release artifacts on git tag push.
- Self-Extracting SFX Installer:** Packaged using 7-Zip LZMA2 ultra compression (-mx9) creating an all-in-one setup binary with archive integrity verification (7z t).

5. Verification Metrics & Baseline Integrity

Comprehensive testing confirms 100% test pass rate with zero regression across all analytical components:

Verification Test Suite	Test Count	Status	Key Verification Scope
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Packaging & Installer	6 / 6	PASSED	Scripts, icons, website CTA URL, .gitignore, and health CLI
GitHub Pages Deployment	6 / 6	PASSED	.nojekyll, relative paths, workflow syntax, clean root
Unified FastAPI Contract	19 / 19	PASSED	/predict, /profile, /health, /info schemas and contracts
Operational GIS Dashboard	9 / 9	PASSED	Map initialization, layer toggles, risk panel, CWC telemetry
Dynamic Risk Fusion & Trigger	67 / 67	PASSED	IMD rainfall fetching, antecedent thresholds, fusion engine
Static Inference & Profiler	74 / 74	PASSED	Model A static LSM, feature encoders, state coordinate bounding
Total Test Suite	181 / 181	PASSED (100%)	Zero regressions across all scientific and UI tiers